

Interpretable point cloud classification using multiple instance learning

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Abstract

Understanding 3D cell shape is crucial in biomedical research, where morphology serves as a key indicator of disease, cellular state, and drug response. However, existing 3D point cloud classification models often lack interpretability, making it difficult to extract biologically meaningful insights. To address this, we propose POINTMIL, an inherently interpretable point cloud classifier using Multiple Instance Learning (MIL). Unlike other methods that rely on global interpretations, POINTMIL simultaneously improves accuracy of point cloud-based classifier backbones and provides fine-grained, point-specific explanations, pinpointing the most informative regions of 3D shapes, without requiring post-hoc analysis. We demonstrate POINTMIL on two publicly available datasets showing state-of-the-art mACC (97.3%) and F1 (97.5%) on the Intra biomedical dataset. Additionally, we introduce a novel dataset of drug-treated cancer cells (Morph3DCell), to show how POINTMIL reveals the morphological effects of drug treatments at a fine-grained level, with implications for drug discovery and mechanism-of-action prediction. Beyond biomedical applications, we show that POINTMIL also offers quality interpretations and improves the classification accuracy on standard shape benchmarks such as ModelNet40 and ScanObjectNN, demonstrating its generalisation to broader 3D object recognition tasks.

1. Introduction

Three-dimensional (3D) imaging data is essential across a range of scientific and industrial applications, including autonomous driving, augmented reality, robotics, and biomedical research. In biomedical imaging, understanding the 3D morphology of cells is increasingly recognised as a powerful approach for characterising cellular states, disease phenotypes, and responses to perturbations such as drug treat-

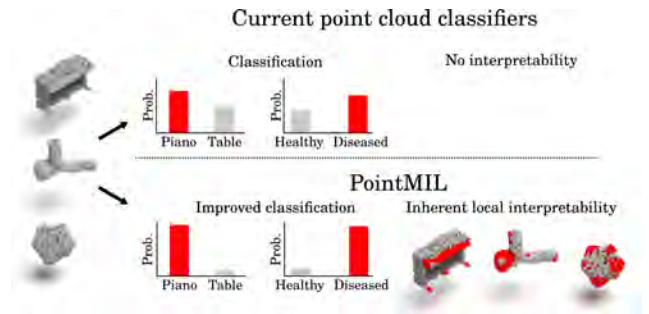


Figure 1. Current point cloud classifiers usually only provide predictive probabilities. We propose POINTMIL to inherently incorporate interpretability and improve predictive performance into point-based architectures.

ments [49]. The ability to systematically classify and interpret changes in 3D cell shape has implications for disease diagnostics [43], biological functions, and drug discovery [9, 11, 20, 24, 30, 49].

Recent advances in point cloud classification have focused mainly on improving model accuracy [19], with limited emphasis on interpretability. However, for scientific applications such as classifying diseased cells and morphological profiling for drug discovery, interpretability is crucial. The morphology of drug-treated cells encodes valuable information about drug mechanisms of action and cellular responses, yet existing models do not highlight where these morphological differences are most informative. Without interpretability, such models function as black boxes, limiting their utility for biomedical researchers who seek to understand how and why a model classifies certain cell shapes under different perturbations.

To address this, we propose POINTMIL, an inherently interpretable point cloud classification framework that provides fine-grained, *local* explanations of classification decisions. Our model is particularly suited for biological applications where local morphological features—such as pro-

trusions, flattening, or shape irregularities—are discriminative for classification. Using a Multiple Instance Learning (MIL) framework [10], our method ensures that each classification decision is based on identifiable, interpretable features within the cell shape. We apply POINTMIL to a novel dataset of cancer cells treated with different drugs, demonstrating how our approach can localise shape regions that are most informative of drug effects. This ability to pinpoint morphological features of drug response has the potential to improve high-throughput screening strategies and advance the use of AI in drug discovery pipelines. Our main contributions are:

- We propose POINTMIL, a MIL framework that provides inherent *local* interpretability for 3D point cloud classification.
- We introduce a case study on classifying cancer cells treated with different drugs, demonstrating that POINTMIL can reveal shape features that are most informative of drug effects.
- We publicly release a novel dataset of point-cloud representations of drug-treated cancer cells to serve as a benchmark for interpretable 3D point cloud classification in biomedical research.
- We validate POINTMIL on other publicly available biomedical datasets, achieving state-of-the-art (SOTA) interpretability and classification performance.
- We demonstrate the versatility of POINTMIL beyond biomedical datasets to standard point cloud classification benchmarks, ModelNet40 [54] and ScanObjectNN [47] showing comparable performance with SOTA.

2. Related work

Point cloud analysis: One of the first methods that used unordered point clouds directly for classification and segmentation was PointNet [35]. PointNet, however, ignored local relationships between points. Subsequently, PointNet++ [36] introduced hierarchical feature learning to capture locality recursively. Many modern algorithms are built on the design philosophy of PointNet++, including convolutional kernel-based [28, 46, 53], graph-based [50, 52, 56], MLP-based [7, 33], and transformer-based methods [1, 5, 18, 59, 61, 62]. Although significant progress has been made in advancing classification and segmentation accuracy, little work has focused on interpretability.

Interpretability on point clouds: Interpretability methods can be classified along two key dimensions: (1) the stage at which interpretability is introduced and (2) the scope of the explanations provided. Regarding the stage, methods are either *post-hoc* or *inherently interpretable*. *Post-hoc* methods generate explanations after the model has made its predictions, often through additional analysis, approximation techniques, or assessing gradients with respect to the input [64]. In contrast, *inherently interpretable* meth-

ods are designed to integrate interpretability into the model itself, producing explanations as part of the prediction process. With respect to scope, methods are categorised as either *local* or *global*. *Local* approaches focus on explaining individual predictions, offering insights specific to a single input. *Global* approaches aim to provide a holistic understanding of the model’s behaviour across all inputs. Since PointNet ++ [36], many point-based models have used some form of sampling and grouping [18, 33, 55, 62], thus losing point-level information in the classification stage. Therefore, most *local* interpretability methods for point cloud classification are *post-hoc*, including gradient-based [21, 60] and surrogate models [45] based on LIME [38]. Zhang et al. [60] and Huang et al. [21] developed explainability methods for PointNet using global average pooling (GAP) and class activation maps. Taghanaki et al. [44] introduced a module into point set encoders that masked points with negligible contributions, leaving only informative points in the classification layer. Similarly, Zheng et al. [63] obtained saliency maps by shifting points to the object centroid and calculating the corresponding loss gradient with respect to the shifted points. However, *post hoc* methods have been shown to be deceptive and often troublesome [16, 26, 39]. For example, the interpretations of *post hoc* methods can differ depending on the interpretability methods [27], leading to convincing but conflicting interpretations for the same classification. *Post-hoc* methods also involve an additional modelling step, raising further concerns about the precision of their interpretations [15]. Few inherently interpretable methods for point cloud classifications have been proposed, and of these, most are *global*. [3] developed XPCC, a prototype-based interpretable model that used point cloud representation distributions to learn class-specific prototypes. Similarly, [16] developed Interpretable3D, a prototype-based global interpretability model that can be used in conjunction with other model architectures for classification and segmentation. However, none of these inherently interpretable methods offers local interpretations on a point-level basis. While global interpretability provides valuable insights into the overall behaviour of a model, local methods can be especially beneficial when understanding specific, individual predictions is crucial, offering more granular and context-sensitive explanations. To our knowledge, no one has yet offered an inherently *locally* interpretable model for point cloud classification. POINTMIL utilises MIL to offer an inherently *locally* interpretable model.

Multiple instance learning: In the typical binary MIL problem, a bag is labelled positive if and only if at least one of its instances is labelled positive [10]; however, there is no access to individual instances during training. MIL algorithms then attempt to classify entire bags of instances and often pinpoint important or class conditional discrimina-

tory instances as interpretability output. Many MIL methods have been proposed for drug activity prediction [10], video image analysis [2], and cancer detection and subtyping [17, 22, 32, 41]. Recently, [14] extended MIL to time series classification in an interpretable plug-and-play framework. However, to our knowledge, no one has used MIL for interpretable point cloud classification.

3. Methods

Given a point cloud $\mathbf{P} \in \mathbb{R}^{N \times 3} = \{\mathbf{p}_i | i = 1, \dots, N\}$, consisting of N points in Cartesian space (x, y, z) , and their associated d -dimensional point features (often point normals, however, these can be the point coordinates if no point-level features exist) $\mathbf{F} \in \mathbb{R}^{N \times d_{in}} = \{\mathbf{f}_i | i = 1, \dots, N\}$, traditional point-based methods use a point-based encoder f_{enc} to learn a global representation $\mathbf{z} \in \mathbb{R}^d$ for $\mathbf{P}$ by aggregating the points with equal weighting (often through adaptive pooling), followed by a classification head f_{clf} .

We propose a new approach by learning a representation $\mathbf{z}_i \in \mathbb{R}^d$ for each point $\mathbf{p}_i$ for $i \in \{1, \dots, N\}$, and then applying MIL pooling for simultaneous classification and interpretability. Our framework consists of a point-based feature extractor f_{enc} and a MIL pooling module f_{MIL} .

3.1. Feature extractor

To develop a point-level feature extractor, we follow much of the Transformer block from [59]. However, unlike [59], we did not use point sampling strategies. Furthermore, we did not use their multi-graph reasoning. This feature extractor aimed to incorporate contextual information into the point cloud features by: (1) grouping points with k -Nearest Neighbours (k -NN), (2) including relative positional embeddings, and (3) refining point-level features through an attention mechanism. These are detailed in Appendix A.

We also presented analysis on PointNet [35], DGCNN [52], CurveNet [55], and PointMLP [33] feature extractors. For PointNet and DGCNN we replaced the classification heads of these architectures with MIL pooling described in Section 3.2. CurveNet and PointMLP downsample the original point cloud. In order to retain point-level features for every point, we slightly adapted these architectures to remove point sampling. We show the affect of this adaptation on classification results so that any difference in performance can then be attributed to the MIL pooling instead of this adaptation. We used PointMLPElite for our analysis. These adaptations are discussed further in Appendices B and C. Each feature extractor produced d -dimensional point-level features $\mathbf{Z} \in \mathbb{R}^{N \times d} = f_{enc}(\mathbf{P})$, for N points which were fed into different MIL pooling algorithms.

3.2. MIL pooling

After obtaining feature representations $\mathbf{z}_i$ for each point $\mathbf{p}_i$, we evaluated four MIL pooling methods that offer inher-

ent interpretability, Instance [51], Attention [22], Additive [23], and Conjunctive [14].

Instance pooling predicts the label of each point through an instance classifier and then pools the predictions by taking the mean:

$$\begin{aligned} \hat{\mathbf{y}}_i &\in \mathbb{R}^c = f_{clf}(\mathbf{z}_i); \\ \hat{\mathbf{Y}} &= \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{y}}_i. \end{aligned} \quad (1)$$

where c is the number of classes.

Attention pooling calculates the attention weights of the point features through an MLP, calculates a weighted average feature representation for the point cloud using those weights and then classifies that features using an MLP:

$$\begin{aligned} a_i &\in [0, 1] = f_{attn}(\mathbf{z}_i); \\ \hat{\mathbf{Y}} &= f_{clf} \left(\frac{1}{N} \sum_{i=1}^N a_i \mathbf{z}_i \right). \end{aligned} \quad (2)$$

Additive pooling calculates attention weights for each point feature, then classifies each point according to its weighted feature vector, and finally produces a bag classification from the mean of all weighted instance classifications:

$$\begin{aligned} a_i &\in [0, 1] = f_{attn}(\mathbf{z}_i); \\ \hat{\mathbf{y}}_i &= f_{clf}(a_i \mathbf{z}_i); \\ \hat{\mathbf{Y}} &= \frac{1}{N} \sum_{i=1}^N \hat{\mathbf{y}}_i. \end{aligned} \quad (3)$$

Conjunctive pooling trains the point attention and point classification heads independently so that attention weights and point predictions are computed on the features alone. The final point cloud classification is given by the weighted sum of the point classifications weighted by the attention weights:

$$\begin{aligned} a_i &\in [0, 1] = f_{attn}(\mathbf{z}_i); \\ \hat{\mathbf{y}}_i &= f_{clf}(\mathbf{z}_i); \\ \hat{\mathbf{Y}} &= \frac{1}{N} \sum_{i=1}^N (a_i \hat{\mathbf{y}}_i). \end{aligned} \quad (4)$$

3.3. Contextual attention

As [14] showed that these pooling operations often produced sparse explanations which occasionally did not cover the entire discriminatory regions, we propose injecting a contextual prior into our calculation of attention, following ideas similar to [17]. For attention-based pooling methods, Attention, Additive, and Conjunctive, attention weights for each point are calculated as:

$$a_i \in [0, 1] = f_{attn}(\mathbf{z}_i), \quad (5)$$

Table 1. Interpretability results in terms of AOPCR and NDCG@n (AOPCR/NDCG@n) on Intra. The best results are given for each method in **bold**.

	PointNet	DGCNN	CurveNet	Transformer
PSM	0.579/0.243	0.916/0.248	1.371/0.218	6.518/0.320
CLAIM	0.967/0.187	6.033 /0.480	1.363/0.252	14.023/0.593
Add.	0.792/ 0.254	4.486/ 0.482	0.615/ 0.266	18.162 / 0.613
Att.	0.005/0.222	−0.031/0.223	1.520/0.260	14.541/0.539
Conj.	0.741/0.208	4.828/0.467	2.660 /0.207	16.305/0.610
Inst.	0.973 /0.225	5.212/0.462	1.709/0.236	16.166/0.587

where f_{attn} is an MLP and $\mathbf{z}_i$ is a feature vector for each point $\mathbf{p}_i$. We propose updating these attention weights according to the attention weights of the nearest neighbours of each point i , such that:

$$a_i^{\text{new}} \in [0, 1] = \frac{1}{k} \sum_{j \in \mathcal{N}(\mathbf{p}_i)} a_j, \quad (6)$$

where $\mathcal{N}(\mathbf{p}_i)$ represents the set of points in the neighbourhood of $\mathbf{p}_i$. This update mechanism smooths the attention weights by incorporating the information from the local neighbourhood, thus addressing the sparsity of the original attention mechanism and providing a more context-aware attention distribution across the point cloud.

3.4. Interpretability

Interpretations were derived through MIL pooling. The Instance pooling strategy classifies each point individually before pooling, yielding point-level predictions: $\{\hat{\mathbf{y}}_i | i = 1, \dots, N\}$. Additive and Conjunctive also make point-level predictions; however, the interpretations are scaled by attention weights: $\{a_i \hat{\mathbf{y}}_i | i = 1, \dots, N\}$. For each of these pooling algorithms, we applied a softmax operation over the class dimension and took the index of the class for which we wished to obtain interpretations, so that we obtained a scalar for each point in the point cloud. For the Attention pooling strategy, we used the attention weights: $\mathbf{a} \in \mathbb{R}^{1 \times N} = \{a_i | i = 1, \dots, N\}$, which were interpreted as a measure of general importance for each point in the point cloud and were not class-specific.

4. Experiments

We compared the interpretability of POINTMIL with other *locally* interpretable point cloud classification methods including class attentive interpretable mapping (CLAIM; [21]), and point cloud saliency maps (PSM; [63]). Similarly to class activation maps (CAM; [64]), CLAIM uses global average pooling (GAP) after point-level feature extractors (the original paper focused on PointNet) and projects the weights of the classifier after GAP on the features of each point to obtain interpretations for each point. PSM assigns

scores to each point based on its contribution to the classification loss. This is done by shifting the points towards the centroid of the point cloud and then calculating the gradient of the loss with respect to each point in spherical coordinates. Interpretability should enhance classification, and not hinder it. To showcase this, we compared POINTMIL to several other point-based architectures in terms of classification performance on six separate datasets, three 3D biological cell-shape datasets the 3D shape classification benchmarks.

4.1. Evaluation metrics

We used the area over the perturbation curve to random (AOPCR; [40]) and normalised discounted cumulative gain at n (NDCG@n) to quantitatively evaluate interpretability [13, 14]. Please see Appendix D for more details. For classification, we used the overall accuracy (oACC), mean class accuracy (mACC), and the F1 score.

5. Datasets

We evaluated POINTMIL on several open source datasets, including two real-world datasets of 3D cell shapes (Intra [58] and 3D red blood cell (RBC) dataset [42]) and two of everyday objects (ModelNet40 [54] and ScanObjectNN [47]). We consider the hardest perturbed variant (PB-T50-RS) of ScanObjectNN in our experiments. See Appendix E for more details.

5.1. Morph3DCell

The phenotypic effects of many drugs remain poorly understood, particularly in 3D [9]. Although some treatments induce recognisable shape changes, the specific morphological features that distinguish drug responses are often unknown. Unlike datasets with defined interpretability ground truth (Intra), or very specific shape classes (RBC), drug-treated cells present a unique challenge: Can our model classify treatments while revealing biologically meaningful shape differences? Answering this could help identify novel mechanisms of action (MOA) and off-target effects by linking specific shape changes to cellular pathways [4, 8, 25, 49].

To further advance research in interpretable 3D point cloud classification for biomedical applications, we introduce the **Morph3DCell**, a real-world dataset of 1500 3D drug-treated cancer cells imaged in 3D using oblique-plane light-sheet microscopy [12, 34]. Cells were treated with Nocodazole, and Blebbistatin, which target microtubules, non-muscle myosin II, respectively, and a control (No Treatment). This dataset serves as a benchmark for studying how drug treatments alter 3D cell morphology. We make this data publicly available to facilitate further research in drug-induced 3D cellular phenotypes. Further information on Morph3DCell is available in Appendix F.

6. Results

6.1. Intracranial vessels

Table 1 shows the interpretability results on the Intra dataset for PointNet, DGCNN, CurveNet and the Transformer backbone. POINTMIL provided better interpretability performance than both PSM and CLAIM, overall. Across backbones, POINTMIL had the highest AOPCR and NDCG@n. The only exception was CLAIM that had a higher AOPCR for the DGCNN backbone. Among the interpretability methods, the Transformer produced the highest AOPCR and NDCG@n results. This could be due to the attention mechanisms within the Transformer block that already enabled the model to focus on informative points, which is further exacerbated by the MIL pooling. Among all backbones, PointNet performed the worst, suggesting that PointNet is not adequate in capturing discriminative morphological cues.

Visualisations of the interpretability for each pooling method on the annotated *Aneurysm* class using the Transformer backbone are shown in Figure 2. The red points indicate areas deemed significant by the model for that specific class. *Aneurysm*'s are presented by the abnormal bulging or ballooning of blood vessels. The first column in Figure 2 shows local annotations of *Aneurysms*, with each other column presenting interpretations for the *Aneurysm* class using the different methods. The last columns show the perturbation curves. These show the decay in the logit of the predicted class after removing the most important points. A larger decay suggests that those points are indeed discriminative for the class. POINTMIL is clearly able to localise on informative regions better than other methods as seen by the visualisation as well as a larger decay in logits shown by the perturbation curve.

Among all MIL pooling methods, Additive and Conjunctive performed best on the Intra dataset. This superior performance of Additive and Conjunctive pooling can be attributed to their ability to better aggregate point-level importance scores. Additive pooling scales point features with their importance weights, preserving detailed information while focusing on relevant points before being passed into a point-level classifier. Conjunctive pooling further enhances this by independently computing attention weights and class-specific contributions, explicitly aligning the model's focus with the predicted class. In contrast, Instance pooling lacks this importance weighting, and Attention pooling does not offer class-specific explanations and rather provides a general measure of importance across classes, which limits their interpretability.

Interpretability should promote classification accuracy and not hinder it. POINTMIL outperformed all methods on Intra in terms of mACC and F1 score by a considerable margin of at least 4.5%. POINTMIL achieved SOTA

on Intra with an mACC of 97.3% and an F1 of 97.5% using Conjunctive pooling with the Transformer backbone (Table 2). Importantly, POINTMIL increased the performance of all the backbones (shown in violet in Table 2).

6.2. Red blood cells

We also present local interpretations for other datasets that lack ground-truth annotations, yet are grouped into classes based on their shapes. Figure 3 illustrates the visual interpretations of POINTMIL with the Transformer backbone for six of the nine classes of RBC with their corresponding perturbation curves. This demonstrates that POINTMIL successfully localises on biologically relevant structural areas. For example, *Discocytes* are characterised by their biconcave shapes, with interpretations for this class focussing on regions identified around the central concavity. In the case of *Acanthocytes*, which exhibit several spicules of varying sizes that project from their surfaces at irregular intervals, POINTMIL similarly focused on these projections to identify this class. For *Knizocytes*, which have a triangular morphology, the model highlighted the areas where the lobes converge. Additionally, POINTMIL pinpointed the spiky projections of *Echinocytes* and *Keratocytes*, as well as the interaction zones where two cells meet in *Cell Clusters*.

Again, POINTMIL outperformed all methods across metrics by up to 11.3% in terms of mACC on RBC.

6.3. Drug-treated cancer cells

We applied our method to a dataset of 1500 melanoma cells imaged in 3D using oblique-plane light-sheet microscopy [12, 34]. Cells were treated with Nocodazole, and Blebbistatin, which target microtubules, non-muscle myosin II, respectively, and a control (No Treatment).

Figure 4 shows the interpretation outputs on the Morph3DCell dataset using POINTMIL with the Transformer backbone. Each row corresponds to the ground truth treatment, while each column represents the interpretation for a given predicted class. The highlighted regions indicate the most influential morphological features driving the classification decision (No Treatment: green, Nocodazole: purple, Blebbistatin: red). Notably, for cells treated with Blebbistatin, elongated extensions are highlighted, aligning with its known disruption of actomyosin contractility through non-muscle myosin 2 inhibition [29]. In contrast, Nocodazole-treated cells, which exhibit more rounded morphologies due to microtubule depolymerization [6], show importance regions across the periphery, suggesting that microtubule organisation plays a key role in model predictions. Across cells, interpretations for Blebbistatin focussed on the protrusions or blebs, while those for Nocodazole focussed on rounder regions. No Treatment interpretations focussed

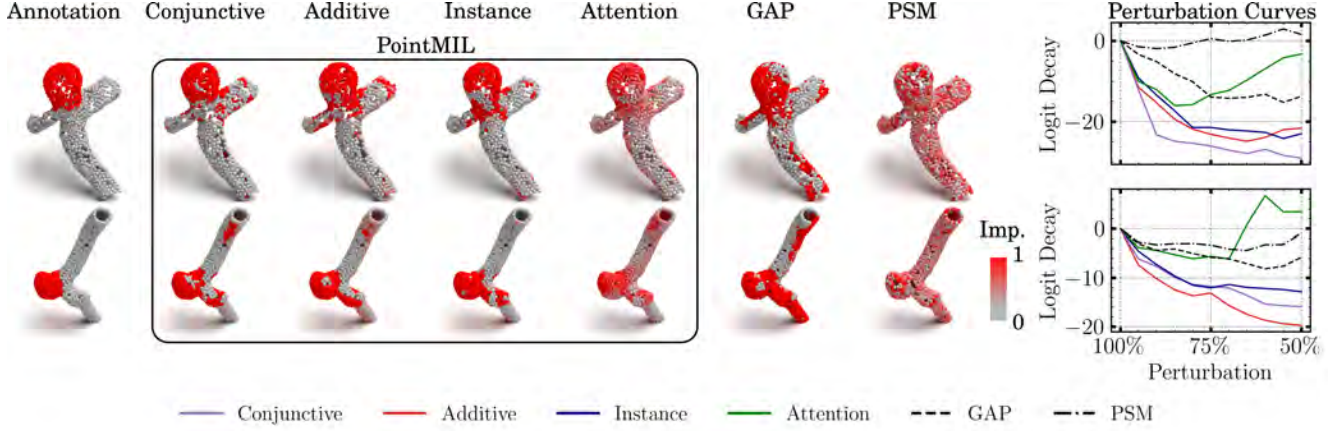


Figure 2. POINTMIL, CLAIM and PSM interpretability visualisations and corresponding perturbation curves using the Transformer backbone for example cells from the IntrA dataset.

Table 2. Classification results on IntrA, RBC, and ModelNet40. All results are shown without voting strategy on 1024 points. The highest results are shown in **bold**. Differences between backbones and POINTMIL are shown in **violet**. Adapted architectures without farthest point sampling results are shown with a †.

Method	Morph3DCell		IntrA		RBC		ModelNet40		ScanObjectNN	
	mACC(↑)	F1(↑)	mACC(↑)	F1(↑)	mACC(↑)	F1(↑)	mACC(↑)	oACC(↑)	mACC(↑)	oACC(↑)
PointNet[35]	79.1	78.8	81.8	82.4	67.7	67.1	86.2	89.2	63.4	68.2
PointNet++[36]	82.4	81.5	92.7	94.2	86.2	87.1	-	91.9	77.6	79.1
PointConv[53]	79.5	79.3	83.0	82.1	68.1	67.9	-	92.5	-	-
DGCNN[52]	80.5	80.2	90.6	91.8	84.8	85.1	90.2	92.9	-	86.0
PCT[18]	77.0	75.9	69.2	68.9	68.7	69.2	-	93.2	-	-
CurveNet[55]	81.3	80.9	88.3	89.8	88.3	87.8	-	93.8	-	-
CurveNet†	83.6	83.1	87.8	87.8	85.8	85.7	90.6	93.4	-	-
PointMLP[33]	80.1	79.4	88.4	88.8	91.8	92.2	91.3	94.1	83.9	85.4
PointMLPElite	81.3	80.0	86.2	86.8	84.7	83.8	90.9	93.6	-	-
PointMLPElite†	80.2	80.4	89.9	89.7	84.2	83.3	90.1	92.6	81.8	83.8
PointNetXt[37]	83.6	83.1	91.8	94.7	86.1	87.1	90.8	93.2	85.8	87.7
3DMedPT[59]	81.8	81.2	92.2	93.3	81.3	83.2	-	93.4	-	-
POINTMIL(PointNet)	82.2+3.1	81.1+2.3	82.0+0.2	82.4+0.0	69.0+1.3	69.1+2.0	87.1+0.9	90.7+1.5	70.4+7.0	73.8+5.6
POINTMIL(DGCNN)	85.1+4.6	84.2+4.0	95.2+3.2	94.6+2.8	92.4+7.6	92.4+7.3	90.8+0.6	93.1+0.2	83.0	85.6-0.4
POINTMIL(CurveNet†)	83.1-0.5	82.9-0.2	91.3+3.5	89.9+2.1	91.2+5.4	90.5+4.8	91.0+0.4	93.5+0.1	-	-
POINTMIL(PointMLPElite†)	82.2+2.0	81.1+0.7	89.9+0.0	91.1+1.4	92.1+7.9	94.3+11.0	90.5+0.4	93.5+0.9	81.4-0.4	83.0-0.8
POINTMIL(Trans.)	84.0+2.2	83.9+2.7	97.3+5.1	97.5+4.2	92.6+11.3	92.2+9.0	89.0	92.7-0.7	80.7	83.2

on areas without these distinct abnormalities.

To quantify interpretability, we performed perturbation analysis, systematically removing the most important points for the correctly predicted class and tracking the change in classification logits (rightmost column of Figure 4). Interestingly, removing the most important regions for both Blebbistatin (red line) and Nocodazole (purple line) cells caused the predictions to turn more towards No Treatment (green line). As expected, random perturbations result in minimal logit change (black dotted line), further supporting the specificity of our importance maps. These results suggest that POINTMIL not only classifies drug treatments but also reveals distinct and biologically meaningful shape changes, which could aid in identifying novel drug effects or off-target interactions in large-scale screens of perturbations with unknown effects.

6.4. Point cloud classification benchmarks

Finally, we evaluate our approach on datasets outside the realm of cell biology to further challenge its versatility. Figure 5 presents the visual interpretations of POINTMIL applied to the ModelNet40 dataset, showcasing a subset of classes. For example, when classifying a *Piano*, the model focused primarily on the keys, while it emphasised the branches and foliage of a *Plant*. The *Bookshelf* displayed red points along the shelves. Similarly, for the *Chair*, crucial features included seat and legs, while the wings and fuselage were highlighted for *Airplane*. More examples are given in Appendix G.

Figure 6 shows the effect of removing the top 10% to 50% of important points on a *Piano* and *Chair* on the logits of those classes. The perturbation curves illustrate that when the points identified as most informative for classi-

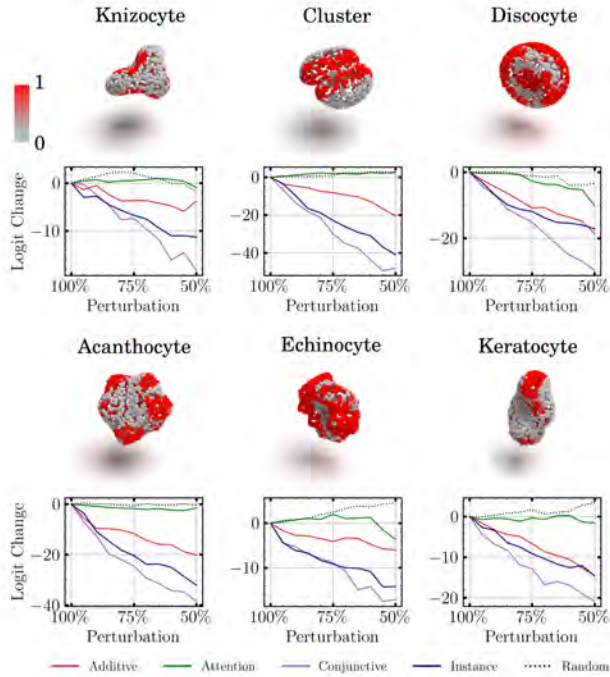


Figure 3. Interpretability visualisation (top row) and corresponding perturbation (bottom row) curves for different RBC shapes.

ying a *Piano* are removed, POINTMIL misclassified the object as a Night Stand. Similarly, when the points identified as the most informative for classifying a *Chair* are removed, POINTMIL misclassifies the object as a TV stand. These interpretations reveal how POINTMIL effectively identified and localised relevant features across various object categories, enhancing our understanding of the model’s decision-making process. Appendix H discusses the different backbones on interpretation outputs.

We additionally assessed POINTMIL on the hardest perturbed variant of ScanObjectNN [47]. POINTMIL achieved results comparable with state-of-the-art. POINTMIL improved accuracy significantly on the PointNet backbone. Although, there was a slight decrease in performance for the DGCNN and PointMLP backbones, these are within the standard deviations reported in the original papers (± 0.5 in terms of oACC for DGCNN [shown in [37]], and ± 0.6 and ± 0.8 in terms of oACC and mACC for PointMLPElite, respectively).

7. Ablation studies

We evaluated the effect of including contextual attention in our attention-based pooling mechanisms: Additive, Attention, and Conjunctive and the impact of varying the value of k (Figure 7). A value of $k = 0$ represented no contextual attention. Including contextual attention consistently offered advantages across all pool-

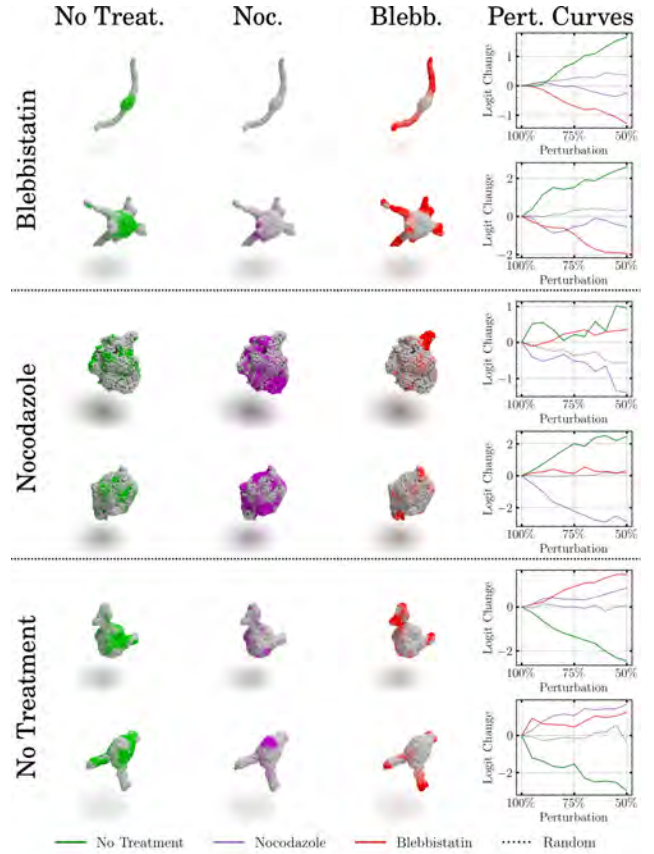


Figure 4. Interpretation outputs on the Morph3DCell dataset using POINTMIL. Each row corresponds to the ground truth drug treatment class (Blebbistatin, Nocodazole, or No Treatment), while each column represents interpretations for different predicted classes. Coloured regions on the 3D cell structures highlight points identified as most important by POINTMIL for classification into the corresponding column’s label. The final column displays perturbation curves, illustrating the logit change in class predictions as the most important points for the correctly predicted class are removed.



Figure 5. Interpretability outputs of PointMIL for different shape classes from ModelNet40

ing methods and metrics compared to not using it. In terms of F1 and mACC contextual attention led to improved performance, particularly with the Conjunctive and Attention mechanisms, which consistently outper-

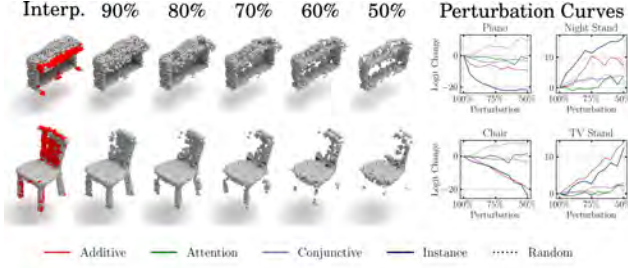


Figure 6. Interpretability outputs and perturbation curves of POINTMIL with the Transformer backbone for different shape classes from ModelNet40

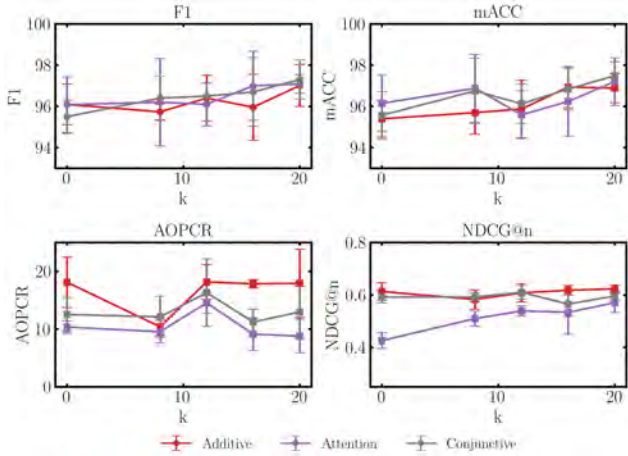


Figure 7. Ablation studies on the value of k in our contextual attention on F1, mACC, AOPCR, and NDCG@n using the transformer backbone.

formed the Additive method as k increased. All pooling methods produced F1 and mACC scores of $> 97\%$ after including contextual attention. For AOPCR, contextual attention was found to be most beneficial when using a value of $k = 12$. Lastly, considering NDCG@n, increasing k provided the most benefit to Attention pooling, while offering slight improvements to Additive and Conjunctive. Additive and Conjunctive pooling outperformed Attention pooling across interpretability metrics, whether or not contextual attention was used. Further results are shown in Appendix I. The choice of pooling method should be guided by the specific requirements of the task and dataset characteristics. For tasks prioritising interpretability, Conjunctive pooling with contextual attention is recommended due to its class-specific focus. For applications prioritising simplicity, Instance pooling offers computational efficiency. An exploration of MIL pooling techniques specific to point cloud data could also enhance this work further. Although contextual pooling improved classification and interpretation

methods, there is a trade-off in computation since the time complexity for k -NN graph search is $O(N^2)$ for the N number of points. The graph construction time complexity is also $O(Nk)$, therefore, as k increases, this process takes longer.

We additionally demonstrate POINTMIL’s robustness to noise. Figure 8 shows how, even when noisy points are added to objects, POINTMIL is still able to focus on salient 3D shape motifs. Further robustness to noise analysis is shown in Appendix J.

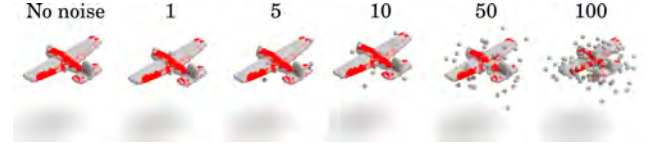


Figure 8. Interpretability visualisations of POINTMIL on a *Airplane* from ModelNet40 after adding a number (shown on the heading) of noisy points. POINTMIL is able to still focus on salient shape motifs ignoring noise.

8. Conclusion

This work introduced POINTMIL, the first framework to apply MIL to point cloud classification, enabling fine-grained, point-specific interpretability without relying on *post-hoc* methods. By integrating MIL-based pooling with a contextual attention mechanism, our approach improves both classification accuracy and interpretability across multiple datasets and architectures.

Our method achieved state-of-the-art performance on the Intra biomedical dataset (97.3% mACC, 97.5% F1-score) and introduced Morph3DCell, a new real-world dataset of drug-treated 3D cancer cells, demonstrating how POINTMIL identifies morphological features linked to drug response. Additionally, we showed that POINTMIL generalises to standard benchmarks such as ModelNet40 and ScanObjectNN, proving its versatility beyond biomedical applications.

By combining interpretability with improved classification, PointMIL advances the field of 3D shape analysis, particularly for biomedical research and morphological profiling. Future work could extend this approach to segmentation-based models for richer point-level representations, explore alternative MIL pooling strategies tailored to point cloud data, and apply POINTMIL to additional datasets with ground-truth interpretability labels. Overall, our framework sets a new standard for interpretable and high-performance 3D classification, offering valuable insights across scientific and real-world applications.

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A. Model details

A.1. Transformer block feature extractor

A.1.1. Group features through k -nearest neighbours:

Formally, we constructed a k -NN graph on $\mathbf{P}$ with the graph including a self-loop to point-level features:

$$\begin{aligned}\mathcal{N}(\mathbf{p}_i) &= \text{KNN}(\mathbf{P}, \|\mathbf{p}_i - \mathbf{p}_j\|_2^2), \mathbf{p}_i, \mathbf{p}_j \in \mathbf{P}, \\ \mathbf{f}'_i &= [(\mathbf{f}_j - \mathbf{f}_i), \mathbf{f}_i]_{j \in \mathcal{N}(\mathbf{p}_i)} \in \mathbb{R}^{k \times 2d_{in}},\end{aligned}\quad (7)$$

where $\text{KNN}(\cdot)$ is the k -NN function, $[\cdot, \cdot]$ is concatenation, k is the hyperparameter of the k -NN graph, $\mathcal{N}(\mathbf{p}_i)$ is the set of neighbours of $\mathbf{p}_i$, and $\mathbf{f}'_i$ is the point feature augmented with local contextual information.

A.1.2. Learned relative positional encoding:

To encode spatial configurations per point-cloud neighbourhood we incorporated positional embeddings, $\mathbf{h}_i$ such that:

$$\mathbf{h}_i \in \mathbb{R}^{k \times d_h} = \phi_{pos}([\mathbf{p}_i - \mathbf{p}_j]_{j \in \mathcal{N}(\mathbf{p}_i)}), \quad (8)$$

where ϕ_{pos} is an MLP and d_h is the output channel dimension of ϕ_{pos} . The features were then further augmented with this positional encoding to give:

$$\mathbf{f}''_i = [\mathbf{f}'_i, \mathbf{h}_i]. \quad (9)$$

Thus, we obtained a new feature set $\mathbf{F}'' \in \mathbb{R}^{N \times k \times (2d_{in} + d_h)} = \{\mathbf{f}''_i\}_{i=1}^N$. This is then passed

A.1.3. Attention on the augmented features:

The resulting features, $\mathbf{F}''$, were then fed into a transformer with EdgeConv as the query operation. Recall that EdgeConv [52] computes graph features for each point using the equation:

$$\mathbf{e}_i \in \mathbb{R}^{d_e} = \max_{j \in \mathcal{N}(\mathbf{p}_i)} (\phi_{edge}(\mathbf{p}_i, \mathbf{p}_j - \mathbf{p}_i)), \quad (10)$$

where ϕ_{edge} is an MLP with output dimension d_e . The $\mathbf{F}''$ were then transformed using attention [48]:

$$\begin{aligned}\mathbf{Q} &\in \mathbb{R}^{N \times d_k} = \text{EdgeConv}(\mathbf{F}'') \mathbf{W}_q \\ \mathbf{K} &\in \mathbb{R}^{(N \times k) \times d_k} = \text{Flatten}(\mathbf{F}'') \mathbf{W}_k \\ \mathbf{V} &\in \mathbb{R}^{(N \times k) \times d_v} = \text{Flatten}(\mathbf{F}'') \mathbf{W}_v,\end{aligned}\quad (11)$$

where $\mathbf{W}_q \in \mathbb{R}^{d_e \times d_k}$, $\mathbf{W}_k \in \mathbb{R}^{(2d_{in} + d_h) \times d_k}$ and $\mathbf{W}_v \in \mathbb{R}^{(2d_{in} + d_h) \times d_v}$ are learnable weight matrices. Our final point-level output features from the transformer block was then given by:

$$\mathbf{z}_i \in \mathbb{R}^{N \times d_v} = \mathbf{q}_i(\text{softmax}(\mathbf{k}_i)^\top \mathbf{v}_i). \quad (12)$$

For all experiments, we used two transformer layers such that the final feature vector for each point was of size 256.

B. CurveNet adaptation

CurveNet uses sampling and grouping. Our only adaptation to CurveNet was use the same number of sampled points as input into the farthest point sampling algorithm. We kept everything else the same as the original paper. We replaced the original adaptive max, adaptive mean pooling, and the classification head with MIL pooling. The final feature vector for each point was of size 1024.

C. PointMLP adaptation

Similar to CurveNet, PointMLP uses sampling and grouping. We adapted PointMLPElite by simply changing the reducers from 2 to 1 to avoid point sampling. We kept everything else the same as the original paper. We replaced the original adaptive max, adaptive mean pooling, and the classification head with MIL pooling. The final feature vector for each point was 256 in size.

D. Interpretability metrics

AOPCR does not require instance labels, whereas NDCG@n does. AOPCR works by removing the most important instances in sequence and observing the impact on prediction accuracy. The faster the prediction declines, the better the ordering, as the most influential instances are removed earlier. When point clouds are annotated, NDCG@n evaluates how closely the model's interpretability ranking matches the true order. It rewards rankings that prioritise relevant instances, with higher scores indicating better alignment and interpretability.

E. Datasets

E.1. Intra

Intra is an open source dataset of 3D intracranial aneurysm [58]. The task is to classify blood vessels as healthy and aneurysm. There is a total of 1909 blood vessel segments, including 1694 healthy vessel segments and 215 aneurysm segments for diagnosis. 116 of the aneurysm segments are expertly annotated. We use Intra to evaluate interpretability, classification, and segmentation.

E.2. Red Blood Cell

We used another dataset of 3D red blood cells (RBC; [42]) for classification. This dataset includes 825 3D red blood cells imaged using confocal microscopy grouped into 9 expertly annotated shape classes. Blood samples were collected from healthy donors and patients using finger prick blood sampling. For inducing RBC shape transitions, blood from 5 healthy donors was treated with NaCl solutions of varying concentrations to create different RBC shapes. Specific shape classes were expertly annotated according to particular motifs. Thus, similar to Intra, RBC was suitable

for evaluating interpretability by the ability to identify these motifs. Segmentation masks are publicly available. We converted the segmentation to mesh objects using marching cubes with Laplacian smoothing, and then sampled points from the vertices of these mesh objects.

E.3. ModelNet40

ModelNet40 [54] is the *de-facto* benchmark for point cloud classification containing 9,843 training and 2,468 testing meshed CAD models belonging to 40 different object classes.

E.4. ScanObjectNN

ScanObjectNN [47] is a point cloud benchmark comprising 15,000 objects categorized into 15 classes, with 2,902 unique real-world object instances. The presence of background clutter, noise, and occlusions makes this benchmark particularly challenging for existing point cloud analysis methods.

E.5. Trainig splits

For IntrA and RBC, we used a five-fold cross-validation and reported the average test metrics across folds. For ModelNet40 and ScanObjectNN, we used the provided train and test splits and reported the test results.

F. Morph3DCell: A new dataset of 3D drug-treated cancer cells in physiologically relevant environments

The dataset consists of 1500 single-cell point clouds, derived from WM266-4 human melanoma cells embedded in tissue-like collagen matrices and imaged in 3D using oblique-plane light-sheet microscopy. Cells were treated with one of the following conditions:

- **Nocodazole (500 cells):** Microtubule inhibitor, causing cells to adopt a rounded morphology.
- **Blebbistatin (500 cells):** Myosin inhibitor, leading to elongated, spindle-like cell shapes.
- **Control (500 cells):** Untreated cells, exhibiting a range of normal human melanoma morphological variability.

The 3D point clouds were generated via the following pipeline: (1) automatic segmentation of individual cells, (2) surface reconstruction using the marching cubes algorithm [31] with Laplacian smoothing, and (3) point cloud sampling of 1024 points per cell from the reconstructed surface mesh. Figure 9 shows renders of the Morph3DCell dataset.

We will publish the single-cell data in three kinds of representations, binary voxel, mesh, and point cloud.

F.1. Dataset Availability and Usage

The dataset will be made publicly available upon acceptance and will be released under the Creative Commons At-

tribution 4.0 (CC-BY 4.0) license. This dataset can be used for:

- Benchmarking point cloud classification models on real-world biological data.
- Studying the 3D morphological effects of small-molecule inhibitors on cancer cells.

A metadata file ('metadata.csv') accompanies the dataset, providing labels and experimental details. Additionally, we provide Python scripts for loading and visualising the dataset.

G. Visual interpretation examples

Figure 10 shows additional interpretability visualisations on ModelNet40.

H. Comparison between backbones

Figure 11 presents the interpretability results for different backbones when classifying a *Bed* with *Additive* pooling (top row) and a *Plant* with *Conjunctive* pooling (middle row) from the ModelNet40 dataset. The perturbation curves are shown in the bottom row. Interestingly, DGCNN, CurveNet, PointMLP, and Transformer backbones consistently highlight similar regions of importance on the *Bed*, particularly focusing on the frame and headboard of the bed, which are key features distinguishing it from other objects. All backbones focussed on the leaves in the *Plant* as opposed to the pot. This consistency across the backbones demonstrates the robustness of POINTMIL in the identification of informative regions. Additionally, the agreement among backbones suggests that POINTMIL effectively leverages the feature representations generated by each model, ensuring the interpretability results are meaningful and aligned with the task.

I. Additional results

This section contains additional results of individual pooling methods.

I.1. Interpretability

Tables 3 show the IntrA interpretability results for each of the pooling methods using the Transformer backbone, showing the effect of contextual attention. The mean and standard deviations on the test sets across the five folds are shown.

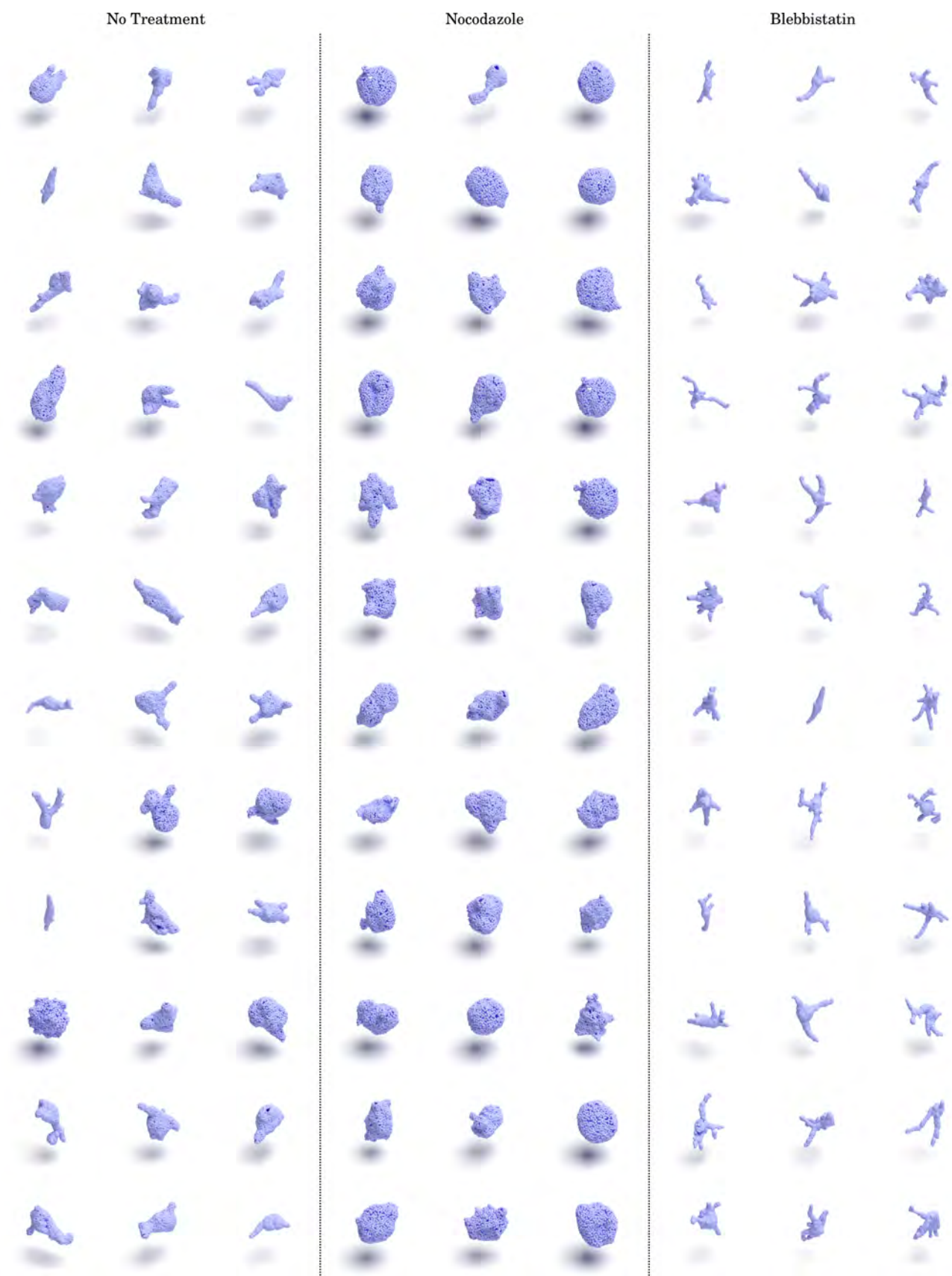


Figure 9. Example rendered point clouds from the Morph3DCell dataset.

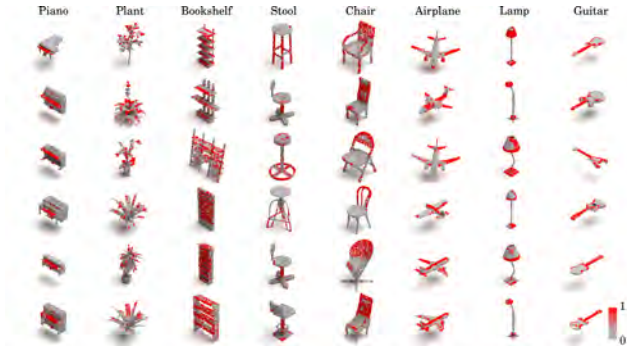


Figure 10. Examples of POINTMIL interpretations for correctly classified shapes from ModelNet40.

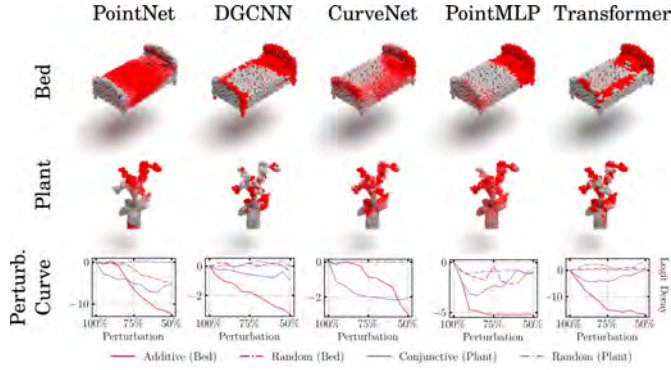


Figure 11. Interpretability of POINTMIL with different backbones on an example *Bed* (top row) and *Plant* (middle row) from ModelNet40. Perturbation curves are shown in the bottom row.

Table 3. Additional POINTMIL interpretability results on Intra using the transformer backbone. We also show the effect of the best contextual attention for each attention-based method.

Model	NDCG@n	AOPCR
Additive	0.613 _{0.033}	18.108 _{4.374}
Additive + context 12	0.608 _{0.035}	18.162 _{3.013}
Attention	0.426 _{0.030}	10.336 _{1.065}
Attention + context 12	0.539 _{0.019}	14.541 _{1.821}
Conjunctive	0.592 _{0.018}	12.526 _{2.960}
Conjunctive + context 12	0.610 _{0.024}	16.305 _{5.859}
Instance	0.587 _{0.022}	16.166 _{3.794}

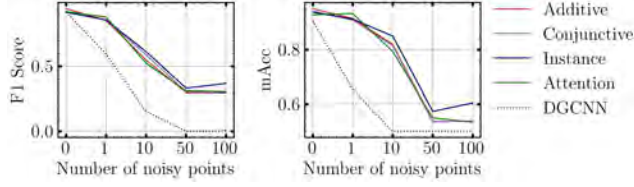


Figure 12. Robustness evaluation of models to noisy inputs.

J. Robustness to noise

Similar to the methods described by Xiang et al. [55] and Yan et al. [57], we assessed the robustness of POINTMIL to noisy inputs. Specifically, we measured the F1 score of models trained on clean (raw) inputs when subjected to noisy inputs during inference. This approach allowed us to evaluate the model’s ability to maintain performance in the presence of input perturbations. The F1 score (left) and the mACC (right) is plotted against the number of noisy points introduced during inference for different POINTMIL methods with the DGCNN backbone and the original DGCNN model in Figure 12. POINTMIL methods demonstrate higher robustness to noise compared to baseline models, with Additive and Conjunctive maintaining consistently higher F1 and mACC scores than the original DGCNN without MIL.

Finally, we demonstrated how POINTMIL could be used to assess where the model went wrong. For example, Figure 13 shows example confusion plots in which the attention of the predicted class is shown in red. Interestingly, for classifying plants, the model only focused on the plant, although when classifying flower pots, the model focused on both the flower and the pot.

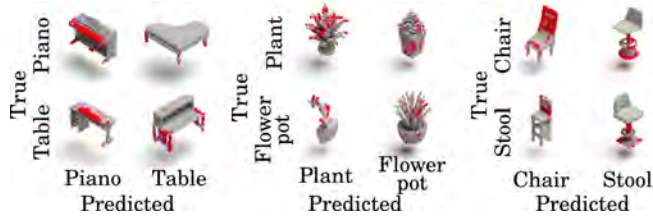


Figure 13. Interpretability visualisations of incorrect classifications from POINTMIL with Transformer backbone on ModelNet40.